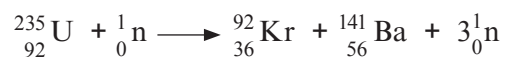


4. (a) A fission nuclear reaction for uranium is:



The masses of the nuclei are:

$$\text{mass of } {}_{92}^{235}\text{U} = 235.01 \text{ u}$$

$$\text{mass of } {}_{36}^{92}\text{Kr} = 91.90 \text{ u}$$

$$\text{mass of } {}_{56}^{141}\text{Ba} = 140.89 \text{ u}$$

$$\text{mass of } {}_0^1\text{n} = 1.01 \text{ u}$$

$$1 \text{ u} = 931 \text{ MeV}$$

Calculate the energy released in this nuclear reaction.

[2]

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- (b) Explain carefully the relevance of the **binding energy per nucleon** curve to nuclear fission and fusion. [6 QER]

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